CURVES AND DIVISORS

In this lecture, we present more facts about algebraic curves, in particular focusing on their divisors. As before, We will not prove all the things covered here as this and the next few lectures are to be taken more like a glossary of terms and results that we will use. The students are expected to work out the details on their own, as this will give them much needed experience with these objects.

1. Local properties of morphisms of curves

We look at the behaviour of a morphism of curves in the neighbourhood of a point.

Definition 1.1. Let $\phi : C_1 \to C_2$ be a nonconstant morphism of smooth curves. Let $P \in C_1$. The *ramification index of ϕ at P*, denoted $e_\phi(P)$, is the quantity

$$e_\phi(P) = \mathrm{ord}_P(\phi^* t_{\phi(P)})$$

where $t_{\phi(P)}$ is a uniformizer at $\phi(P)$. Note that this definition is independent of the choice of uniformizer.

Since ϕ induces a local ring homomorphism between the local rings at P and $\phi(P)$, it is a standard exercise to see that $e_\phi(P) \geq 1$. If $e_\phi(P) = 1$, we say ϕ is *unramified at P*, and if ϕ is unramified at every point on C_1, we say it is an unramified morphism.

Geometrically speaking, choosing a uniformizer at a point is like choosing a local coordinate t on the curve. If $e_\phi(P) = n$, the definition implies that the morphism ϕ looks like the morphism $t \to t^n$ in a neighbourhood of P. If you think of this in terms of a small disk around the origin in the complex plane, you will see that ramified morphisms look like a sector of appropriate size wrapping around the disk and the disk wraps around itself n times.

Proposition 1.2. *Let $\phi : C_1 \to C_2$ be a nonconstant morphism of smooth curves. over an algebraically closed field K*

- *For every $Q \in C_2$,*
$$\sum_{P \in \phi^{-1}(Q)} e_\phi(P) = \deg \phi.$$
- *For all but finitely many $Q \in C_2$,*
$$|\phi^{-1}(Q)| = \deg_s \phi.$$
- *Let $\psi : C_2 \to C_3$ be another nonconstant morphism of smooth curves. Then for all $p \in C_1$,*
$$e_{\psi\circ\phi}(P) = e_\phi(P) e_\psi(\phi(P)).$$

The proof is a standard exercise in local ring theory that you are highly encouraged to write down for yourself.

Corollary 1.3. *A nonconstant morphism $\phi : C_1 \to C_2$ of smooth curves is unramified if and only if*

$$|\phi^{-1}(Q)| = \deg \phi$$

for all $Q \in C_2$.

These are all standard facts about Dedekind domains, really. And this explains the analogous nature of the theory of smooth curves with algebraic number theory.

2. Morphisms in characteristic $p > 0$: A special case

Assume now that $\mathrm{char}(K) = p > 0$ and let $q = p^r$. Recall that K is assumed to be perfect. For any polynomial $f \in K[X]$, let $f^{\mathfrak{q}}$ be the polynomial obtained from f by raising each coefficient of f to the q-th power.

Definition 2.1. For any curve C/K, we define its Frobenius twist curve $C^{(q)}/K$ as the curve whose homogeneous ideal is:

$$I(C^{(q)} = \text{ Ideal generated by } \{f^{(q)} : f \in I(C)\}.$$

There is a natural map defined over K from C to $C^{(q)}$:

$$\phi : C \to C^{(q)}, \qquad \phi([x_0 : \ldots : x_n)] = [x_0^q : \ldots x_n^q].$$

This is called the q-th power Frobenius morphism.

The next proposition enlists basic properties of the Frobenius map. These will be used repeatedly throughout the course.

Proposition 2.2. *Let K be a perfect field with $\mathrm{char}(K) = p > 0$ and let $q = p^r$. Let C/K be a curve and let $\phi : C \to C^{(q)}$ be the q-th power Frobenius morphism.*

- $\phi^* K(C^{(q)}) = K(C)^{\mathfrak{q}} = \{f^q : f \in K(C)\}$.
- *ϕ is purely inseparable.*
- $\deg \phi = q$.

Proof. The first assertion is an easy identification of rational functions using the fact that K is perfect. The second is an immediate consequence of the first. For the third statement, take a finite extension of K if necessary to assume the existence of a smooth point P over K and carry out the degree computation by choosing a uniformizer t at P and writing all the relevant fields as extensions of $K(t)$, using standard facts about separable and purely inseparable field extensions. □

Corollary 2.3. *Every morphism $\psi : C_1 \to C_2$ of smooth curves over a perfect field K of characteristic $p > 0$ factors as*

$$C_1 \xrightarrow{\phi} C_1^{(q)} \xrightarrow{\lambda} C_2$$

where $q = \deg_i \psi$ is the inseparable degree, ϕ is the q-th power Frobenius morphism, and λ is a separable morphism.

This follows from the previous theorem and the correspondence between functions fields and curves.

3. Divisors on curves

A lot of algebraic geometry of a variety is studied by understanding its *codimension* 1 subvarieties: loosely, this means understanding $(n-1)$-dimension subvarieties of an n-dimensional variety. This can get complicated for a general variety, but for curves, since the only codimension 1 subvarieties can be points, this is much more accessible. As usual, this is best understood over algebraically closed fields, and many times we will have that hypothesis in mind even if we don't explicitly say it.

Definition 3.1. The divisor group of a curve C, denoted $\mathrm{Div}(C)$, is the free abelian group generated by the points of C. A divisor $D \in \mathrm{Div}(C)$ is a *formal sum* $D = \sum_P n_P P$, where $n_P \in \mathbb{Z}$ and $n_P = 0$ for all but finitely many $P \in C$. The *degree* of D is $\deg D = \sum_P n_P$ is well-defined. This gives a homomorphism to $\mathbb{Z}$. Its kernel is denoted $\mathrm{Div}^0(C)$.

If C is defined over K, G_K acts on $\mathrm{Div}(C)$ and $\mathrm{Div}^0(C)$ in the obvious way. Then D is *defined over K* if $D^\sigma = D$ for all $\sigma \in G_K$. Note that this does not necessarily mean that each of the constituent points of D are defined over K. We denote by $\mathrm{Div}_K(C)$ and $\mathrm{Div}_K^0(C)$ the corresponding groups of divisors defined over K.

Let C be a smooth curve now. For each function $f \in \bar{K}(C)^*$, we can associate a divisor $\mathrm{div}(f)$ to it by the formula

$$\mathrm{div}(f) = \sum_{P \in C} \mathrm{ord}_P(f) \cdot P.$$

This is a divisor since every nonzero function f has finitely many zeroes and poles. If $f \in K(C)$, then $\mathrm{div}(f) \in \mathrm{Div}_K(C)$ because of how G_K acts on functions.

Since each ord_P is a valuation, $\mathrm{div} : \bar{K}(C^*) \to \mathrm{Div}(C)$ is an abelian group homomorphism. This prompts the following definition(s).

Definition 3.2. A divisor D is called a *principal divisor* if $D = \mathrm{div}(f)$ for some $f \in \bar{K}(C^*)$. Two divisors D_1, D_2 are *linearly equivalent* if $D_1 - D_2$ is principal. We denote by $\mathrm{Pic}(C)$ the *Picard group of* C is the quotient of Div(C) by its subgroup of principal divisors. We define $\mathrm{Pic}_K(C)$ to be its subgroup fixed by the G_K-action. Note that this does not in general equal the quotient of $\mathrm{Div}_K(C)$ by its subgroup of principal divisors defined over K.

Proposition 3.3. *Let C be a smooth curve and let $f \in \bar{K}(C)^*$.*

- $\mathrm{div}(f) = 0$ *if and only if* $f \in \bar{K}^*$.
- $\deg \mathrm{div}(f) = 0$.

The first assertion follows from the fact that if f is a morphism from C to $\mathbb{P}^1$ that has no poles, it is not surjective and hence constant. The second is a standard exercise. In particular, the principal divisors form a subgroup of $\mathrm{Div}^0(C)$.

Definition 3.4. We define the degree 0 part of the Picard group, $\mathrm{Pic}^0(C)$, as the quotient of $\mathrm{Div}^0(C)$ by the subgroup of principal divisors. As usual, we let $\mathrm{Pic}^0_K(C)$ to be the subgroup of $\mathrm{Pic}^0(C)$ fixed by G_K.

The definitions and proposition above is summarized by the following exact sequence:

$$1 \to \bar{K}^* \to \bar{K}(C)^* \xrightarrow{\mathrm{div}} \mathrm{Div}^0(C) \to \mathrm{Pic}^0(C) \to 0.$$

This is the function field analogue of the fundamental exact sequence in algebraic number theory for a number field L, viz.

$$1 \to \text{Units of } \mathcal{O}_L \to L^* \to \text{Fractional ideals of } L \to \text{Ideal class group of } L \to 1.$$

Example 3.5. On $\mathbb{P}^1$, every divisor of degree 0 is principal. How to see this? If $D = \sum n_P P$, with $P = [\alpha_P : \beta_P] \in P^1(\bar{K})$, then the choice of function f vanishing on D is the product $(\beta_P X - \alpha_P Y)^{n_P}$ and $\sum_{n_P} = 0$ ensures that this is actually a well-defined function on homogeneous coordinates. Hence, $\mathrm{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$.